

Measurement of the Linear Thermal Expansion Coefficient of Five Metals by a Capacitance Method

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Abstract

The coefficient of linear thermal expansion α was determined for copper, aluminium, brass, steel, and an unknown metal over the approximate temperature range 20 °C to 70 °C, using a parallel-plate capacitor to detect sub-millimetre rod length changes. A calibration line relating inverse capacitance ($1/C$) to plate separation (Δd) allowed displacement to be inferred directly from capacitance readings. Linear fits to displacement versus temperature rise yielded $\alpha = (18.65 \pm 0.36) \times 10^{-6} \text{ }^\circ\text{C}^{-1}$ (copper), $(21.44 \pm 0.41) \times 10^{-6} \text{ }^\circ\text{C}^{-1}$ (aluminium), $(22.26 \pm 0.91) \times 10^{-6} \text{ }^\circ\text{C}^{-1}$ (brass), $(10.65 \pm 0.16) \times 10^{-6} \text{ }^\circ\text{C}^{-1}$ (steel), and $(20.14 \pm 0.34) \times 10^{-6} \text{ }^\circ\text{C}^{-1}$ (unknown metal). Steel agreed with literature to within 0.3σ ; the remaining metals exhibited systematic offsets of up to 6σ , attributed to a shared systematic source. The grey appearance, density close to steel, and measured α of the unknown metal were consistent with tin ($\alpha_{\text{lit}} = 22.0 \times 10^{-6} \text{ }^\circ\text{C}^{-1}$).

1. Introduction

For most engineering materials, the change in length with temperature can be approximated as a linear relationship across moderate temperature ranges (such as the 20 °C to 70 °C range used in this experiment) [1]. If L_0 is the length of a rod at an initial temperature T_0 , and $\theta = T - T_0$ is the temperature rise, the length at temperature T is

$$L(T) = L_0(1 + \alpha \theta), \quad (1)$$

where α is the coefficient of linear thermal expansion, with SI units of K^{-1} (numerically equal to $^\circ\text{C}^{-1}$). Values of α are material specific and are usually in the range of $4 \times 10^{-6} \text{ }^\circ\text{C}^{-1}$ to $30 \times 10^{-6} \text{ }^\circ\text{C}^{-1}$ for common metals [2]. Knowledge of α is essential in the design of precision instruments, bridges, and pipelines, where differential expansion between materials must be controlled [3]. Over larger temperature ranges, the approximation fails as α is a function of T .

The change in length for a 50 °C rise for a 50 cm rod with $\alpha \sim 20 \times 10^{-6} \text{ }^\circ\text{C}^{-1}$ is only 0.5 mm, which is too small for measurement using a mechanical gauge with suitable precision. For a parallel-plate capacitor with plate area A and plate separation d , the capacitance C is

$$C = \frac{\epsilon_0 A}{d}, \quad (2)$$

because d appears in the denominator, even small changes in separation produce measurable changes in capacitance. This makes the capacitor a sensitive way to detect the small expansion of a metal rod [4]. Rearranging gives $\Delta d \propto (1/C)$, so plotting $(1/C)$ against Δd shows a linear relationship which allows least-squares fitting, whereas C itself varies non-linearly with d .

In this report, the coefficient of linear thermal expansion is determined for copper, aluminium, brass, steel, and an unknown metal by measuring capacitance change as each rod expands, pushing the plates together. The linearity of the expansion is examined, α is determined from the gradient of the fitted line, and the results are compared with the values in the literature.

2. Method

The experimental setup is shown in Figure 1. Each metal rod (≈ 500 mm) was heated using a resistance wire with a current of approximately 1.2 A. A thermocouple was used to monitor temperature, and the capacitance was recorded every 5 °C starting at room temperature up to approximately 70 °C. An adjusting micrometer set the initial separation to the lowest capacitance still on the calibration graph. No potential difference was applied directly across the plates.

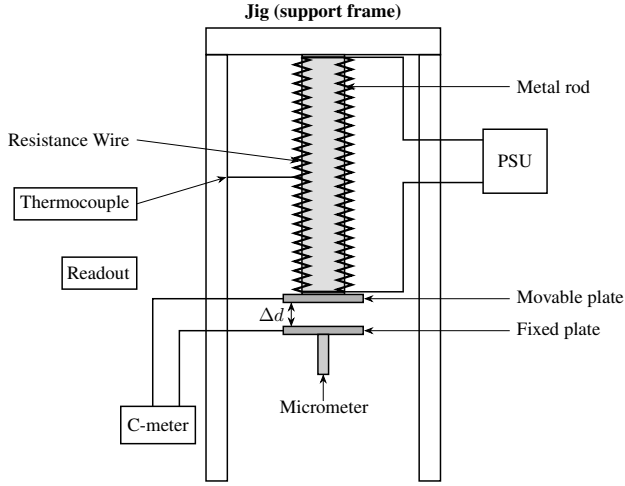


Figure 1: Schematic of the experimental apparatus. The metal rod hangs vertically from the jig, heated by a resistance wire, with a thermocouple attached to its surface. As the rod expands, the movable capacitor plate is pushed down, decreasing separation d and increasing capacitance. The PSU drives current through the heater wire.

2.1 Calibration

Before measuring any thermal expansions, the relationship between plate separation Δd and the inverse capacitance $1/C$ was established by advancing the micrometer in steps of 0.2 mm from $\Delta d = 0$ to $\Delta d = 1.0$ mm, then, back in steps of 0.2 mm offset by 0.1 mm, giving a combined dataset at 0.1 mm intervals across the full range to check for micrometer hysteresis. No significant hysteresis was observed, confirming the calibration relationship is path-independent. The gradient k of the linear fit to $1/C$ against Δd gives

$$\frac{1}{C} = k \Delta d + \frac{1}{C_0}, \quad (3)$$

where $1/C_0$ is the intercept, from the calibration fit. From equation (3), the displacement due to thermal expansion was then found as

$$\Delta d = \frac{1/C - 1/C_0}{k}. \quad (4)$$

2.2 Thermal expansion measurements

The rod was heated by connecting the resistance wire to a DC power supply. Temperature was allowed to stabilise at each interval before the capacitance was recorded, ensuring quasi-static conditions. Data were recorded at 5°C intervals over the approximate range 20°C to 70°C , giving eleven data points per rod. The rod length L_0 at room temperature was measured with a metre rule. The uncertainty in this measurement was estimated to be 0.5 mm (half of one millimetre scale division).

From equation (1), the gradient m of a linear fit to Δd against θ satisfies

$$m = \frac{d(\Delta d)}{d\theta} = \alpha L_0, \quad (5)$$

so that

$$\alpha = \frac{m}{L_0}. \quad (6)$$

Least-squares linear regression (via `linregress` in Python) was applied to each dataset to obtain m and its standard error σ_m . The fractional uncertainty in α was then propagated in quadrature:

$$\frac{\Delta\alpha}{\alpha} = \sqrt{\left(\frac{\sigma_m}{m}\right)^2 + \left(\frac{\Delta k}{k}\right)^2 + \left(\frac{\Delta L_0}{L_0}\right)^2}. \quad (7)$$

3. Results

3.1 Calibration

Figure 2 shows the inverse capacitance $1/C$ as a function of micrometer displacement Δd . The relationship is linear across the full calibration range, confirming that the approximation of equation (3) is valid over the range of displacements used. The fitted calibration gradient was $k = (3.729 \pm 0.046) \times 10^{-3} \text{ pF}^{-1} \text{ mm}^{-1}$. Horizontal error bars are omitted as the micrometer reading uncertainty (0.0005 mm - half of the smallest division) is smaller than the plot markers.

3.2 Thermal expansion

The displacement Δd inferred from equation (4) is plotted against temperature rise θ for each rod in Figure 3. The fractional contribution of the temperature reading uncertainty (0.3%, as quoted in the thermocouple readout specification) to the fitted gradient is $\Delta\theta/\theta_{\text{range}} \approx 0.15^\circ\text{C}/50^\circ\text{C} = 0.3\%$, negligible compared to the gradient standard error of 1–4%, and is therefore omitted from equation (7). For all metals, the relationship was found to be linear over the 50°C range which is consistent with the model of equation (1). Table 1 summarises the fitted gradients, rod lengths, and derived values of α .

The aluminium dataset was collected on two separate occasions. The first dataset (nine points, 20.2°C to 60.2°C) yielded $\alpha = (30.87 \pm 0.77) \times 10^{-6} \text{ }^\circ\text{C}^{-1}$, which is significantly higher than the accepted literature value. The second (complete) dataset (eleven points, 21.5°C to 71.5°C) gave $\alpha = (21.44 \pm 0.41) \times 10^{-6} \text{ }^\circ\text{C}^{-1}$, in closer agreement with literature. As a result of this, the second dataset was used as the primary set of results for aluminium. The difference between the two sets is discussed in Section 4.

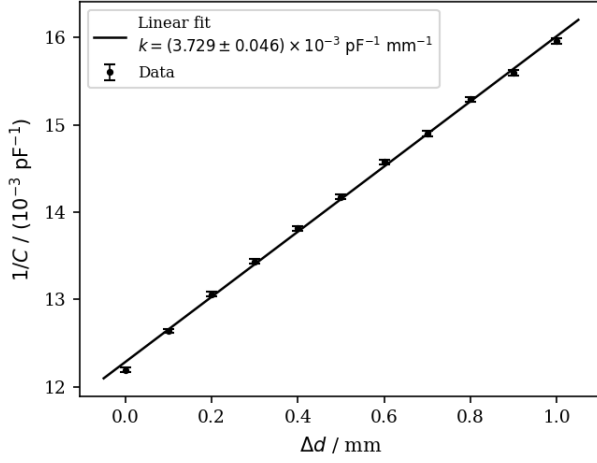


Figure 2: Inverse capacitance $1/C$ as a function of micrometer displacement Δd . Error bars represent the 0.2% capacitance meter uncertainty. The solid line is a least-squares fit with gradient $k = (3.729 \pm 0.046) \times 10^{-3} \text{ pF}^{-1} \text{ mm}^{-1}$.

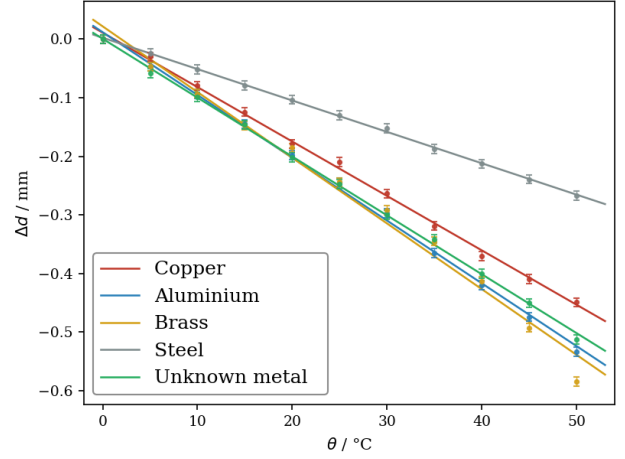


Figure 3: Thermal displacement Δd inferred from equation (4) plotted against temperature rise θ for each metal rod. Error bars represent the 0.2% capacitance meter uncertainty propagated through the calibration. Solid lines are least-squares fits. Fitted gradients and derived values of α are given in Table 1.

Table 1: Fitted gradients m , rod lengths L_0 , and derived coefficients of linear thermal expansion α for each metal. Gradients are negative as Δd decreases during expansion. $\Delta\alpha$ is propagated via equation (7), combining σ_m , calibration gradient uncertainty $\Delta k/k = 1.2\%$, and rod-length uncertainty $\Delta L_0 = 0.5 \text{ mm}$ in quadrature. Literature values are from Kaye & Laby [2].

Metal	$m / (10^{-3} \text{ mm } ^\circ\text{C}^{-1})$	$\sigma_m / (10^{-3} \text{ mm } ^\circ\text{C}^{-1})$	L_0 / mm	$\alpha / (10^{-6} ^\circ\text{C}^{-1})$	$\alpha_{\text{lit}} / (10^{-6} ^\circ\text{C}^{-1})$
Copper	-9.290	0.139	498	18.65 ± 0.36	16.5
Aluminium ^a	-10.72	0.157	500	21.44 ± 0.41	23.1
Brass	-11.22	0.435	504	22.26 ± 0.91	19.0
Steel	-5.359	0.048	503	10.65 ± 0.16	10.7
Unknown metal	-10.05	0.114	499	20.14 ± 0.34	—

^aFirst dataset discarded ($\alpha = (30.87 \pm 0.77) \times 10^{-6} ^\circ\text{C}^{-1}$); see Section 4.

3.3 System resolution

The minimum detectable displacement is limited by the precision of the capacitance meter. With a stated uncertainty of 0.2% and a typical capacitance of 65 pF, the absolute uncertainty in C is approximately 0.13 pF. The corresponding uncertainty in Δd is obtained by propagating through equation (4):

$$\Delta(\Delta d) = \frac{\Delta C}{C^2 k} = \frac{0.13}{65^2 \times 3.729 \times 10^{-3}} \approx 8.3 \mu\text{m}, \quad (8)$$

demonstrating sub-10 μm sensitivity.

4. Discussion

4.1 Comparison with literature

For steel, the measured value $\alpha = (10.65 \pm 0.16) \times 10^{-6} ^\circ\text{C}^{-1}$ agrees with the literature value of $10.7 \times 10^{-6} ^\circ\text{C}^{-1}$ to within 0.3σ . Steel's agreement may be partly attributed to its low thermal conductivity,

which promotes a more uniform temperature distribution along the rod, reducing the difference between the thermocouple reading and the true mean rod temperature. Further, its smaller α produces less total plate displacement, reducing the fractional contribution of any residual tilt.

The deviations for copper (6.0σ), aluminium (4.0σ), and brass (3.6σ) suggest a shared systematic error source. Since α for brass depends on alloy composition, the literature value is not precisely defined, and its larger scatter (gradient standard error roughly three times that of the other metals) may reflect how the alloy composition varied along the rod length producing deviations from a single linear fit. The offsets vary in both sign and magnitude (ranging from -1.7 to $+3.3 \times 10^{-6} ^\circ\text{C}^{-1}$), which is inconsistent with a single offset such as plate tilt. Non-uniform heating along the rod is a more realistic primary source of systematic error since the thermocouple takes readings at a single

point, it could read above or below the true mean rod temperature depending on the local temperature gradient. This differs between metals due to their differing thermal conductivities. Plate tilt provides an additional contribution, and can explain copper’s systematically high value; a tilt of 1° (~ 1 mm edge height difference across the 10 cm plate) is estimated from the plate geometry to produce a roughly 13% overestimation in inferred displacement, consistent with copper’s measured value of α .

The rod-length measurement contributes a 0.1% fractional uncertainty (metre rule, ± 0.5 mm), small compared to the gradient standard error (typically 1–4%), which is the dominant contribution to $\Delta\alpha$ for all metals.

The anomalously high value from the first aluminium dataset ($30.87 \times 10^{-6} \text{ }^\circ\text{C}^{-1}$) cannot be attributed to the same systematic offsets seen across all runs. The most likely cause is the rod shifting within the jig, introducing a sudden displacement unrelated to thermal expansion and inflating the apparent gradient.

4.2 Identity of the unknown metal

The unknown metal had a grey, non-uniform appearance and a measured expansion coefficient of $\alpha = (20.14 \pm 0.34) \times 10^{-6} \text{ }^\circ\text{C}^{-1}$. Table 2 compares this with metals which have similar physical properties.

Table 2: Comparison of the measured expansion coefficient of the unknown metal with literature values for candidate metals [2,3].

Metal	$\alpha_{\text{lit}} / (10^{-6} \text{ }^\circ\text{C}^{-1})$	$ \Delta\alpha / (10^{-6} \text{ }^\circ\text{C}^{-1})$
Silver	18.9	1.2
Tin	22.0	1.9
Nickel	13.4	6.7
Iron	11.8	8.3
Zinc	30.2	10.1

As noted in Section 4.1, the systematic offsets are not consistent in sign or magnitude (ranging from -1.7 to $+3.3 \times 10^{-6} \text{ }^\circ\text{C}^{-1}$), so no reliable correction can be applied. Tin is identified as the most likely candidate: its grey, matte appearance and density (7.3 g cm^{-3} , slightly below steel’s 7.9 g cm^{-3}) matched the rod’s observed colour and weight, and its systematic offset ($-1.9 \times 10^{-6} \text{ }^\circ\text{C}^{-1}$) is close to the lower bound of the range observed across known metals (-1.7 to $+3.3 \times 10^{-6} \text{ }^\circ\text{C}^{-1}$). Although tin melts at 232°C , the experiment reached only 70°C , so this does not preclude its use. Silver, while numerically closer ($|\Delta\alpha| = 1.2 \times 10^{-6} \text{ }^\circ\text{C}^{-1}$), has a density of 10.5 g cm^{-3} and a bright lustre inconsistent with the observed appearance. It is therefore concluded that the unknown metal is most likely tin.

4.3 Suggested improvements

To address non-uniform heating, multiple thermocouples distributed along the rod length would allow the true mean temperature to be measured directly. For plate tilt, a three-point mount with fine adjustment screws would allow for more reliable parallelism than the single micrometer. To reduce the gradient standard error, the temperature range could be extended, and temperature interval decreased.

5. Conclusions

The coefficient of linear thermal expansion was measured for five metals using a capacitive displacement sensor over the approximate range 20°C to 70°C . The thermal expansion was found to be linear with temperature in all cases, consistent with the standard model. The measured values were $\alpha_{\text{Cu}} = (18.65 \pm 0.36) \times 10^{-6} \text{ }^\circ\text{C}^{-1}$, $\alpha_{\text{Al}} = (21.44 \pm 0.41) \times 10^{-6} \text{ }^\circ\text{C}^{-1}$, $\alpha_{\text{Br}} = (22.26 \pm 0.91) \times 10^{-6} \text{ }^\circ\text{C}^{-1}$, $\alpha_{\text{St}} = (10.65 \pm 0.16) \times 10^{-6} \text{ }^\circ\text{C}^{-1}$, and $\alpha_{\text{unk}} = (20.14 \pm 0.34) \times 10^{-6} \text{ }^\circ\text{C}^{-1}$. Steel showed the best agreement with its accepted value. Systematic offsets in the remaining metals, varying in both sign and magnitude, were primarily attributed to non-uniform heating along the rod, with residual plate tilt providing an additional contribution. The unknown metal was identified as most likely tin on the grounds of its expansion coefficient, visual appearance, and density. The system resolution was estimated at approximately $8.3 \mu\text{m}$, demonstrating the suitability of the capacitance method for sensing sub-millimetre thermal displacements.

References

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